ROHAN MALIGE

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Education

University of California, Davis — **September 2021 – June 2025**
Bachelor of Science in Computer Science and Engineering — **Overall GPA: 3.59**
Dean's Honors List (Fall 2021,Spring 2022,Fall 2023,Winter 2025), Edward Frank Kraft Prize (Fall 2021)

Software Skills

Programming Languages: Python, C, C++, Go, Java, Javascript, Swift, HTML, CSS, SQL, MATLAB, cURL, Flutter, iOS.
Frameworks and Libraries: React, React Native, Next.js, Node.js, SwiftUI, GraphQL, Docker, PyTorch, TensorFlow, Keras, Pandas, NumPy, Scikit, Matplotlib, Plotly Dash, Flask, Django, Apple Core Motion, Apple CloudKit.
Databases and Design: AWS DynamoDB, MongoDB, ResilientDB, MySQL, PostgreSQL, Figma.
Cloud and Platforms: Microsoft Azure, GCP, Apple CloudKit, Postman, Firebase Firestore, OpenAI, Engineering Tool Software (ETS), Kubernetes, NERSC Supercomputers.
Technologies and Tools: Git, Machine Learning, Generative AI, Linux, UNIX, Shell scripts, CUDA programming, Wireshark.

Experience

NSF x UC Davis Summer Undergraduate Research Internship — **July 2024 – August 2024**

- Analyzed NASA satellite reflectance data using machine learning models to improve the detection of phytoplankton populations, a key climate change indicator, and refined the estimation of absorption coefficients at specific light wavelengths to more accurately measure their concentrations in ocean waters.
- Compared and fine-tuned multiple ML models (CNNs, MDNs, UNet, 1D CNNs, MLPs) using advanced preprocessing techniques like spectral smoothing and noise reduction to enhance predictive reliability at finer intervals.
- Improved MDN model accuracy by over 15% and increased detection resolution from 5.7nm to 1nm, thereby enabling more precise ocean monitoring and ecological forecasting.

Ben Shalom Lab, UC Davis MIND Institute — **August 2023 – June 2025**
Undergraduate Research Assistant

- Focused on early detection of brain diseases by analyzing individual neuron spikes in lab-grown mice brains.
- Developed a ETL pipeline for neuron-spike analysis, utilized Python's 'multiprocessing' module to optimize the data analysis speed by 80% and transitioned the code base to NERSC supercomputers for further performance improvements.
- Designed a Plotly Dash based web interface allowing users to upload CSV and MATLAB data to generate visualizations like bar plots, histograms, and LDA plots in SVG and PNG formats. Link: **https://github.com/roybens/MEA_Analysis**
- Currently researching machine learning classification algorithms to predict healthy and diseased neuron types.

Projects

UC Davis Capstone Project – Pain & Symptom Tracking App — **January 2025 – June 2025**

- Developed an iOS app for Parkinson's patients by integrating Apple HealthKit and Core Motion, enabling real-time symptom tracking by collecting step count, calories burnt, heart rate variability, and sleep data.
- Leveraged CloudKit for cloud-based storage of raw health data and integrated Auth0 to allow secure access for researchers while upholding HIPAA compliance and patient privacy.
- Designed an accessible UI using Figma, prioritizing simplicity for elderly users with structured workflows, push notifications, and easy data entry.

CrypGo — **September 2023 - November 2023**

- Developed a mobile app wallet called CrypGo using React Native and a blockchain fabric called ResilientDB.
- Utilized API calls directed to GraphQL (https://cloud.resilientdb.com/graphql) through Postman to enable users to access their accounts, submit transactions, and view their activity on the go.
- Outlined enhancements focusing on comprehensive UI development and extending app features under the guidance of Prof. Mohammed Sadoghi.

SacHacks website (Greater Sacramento's premier hackathon) — **June 2024 – February 2025**

- Developed the SacHacks website (https://sachacks.io/) using React.js and CSS and hosted it using Vercel. Incorporated animations to enhance user experience, and a timer to count down to the hackathon event.
- Customized an open-source judging platform for SacHacks and engineered a live score keeping functionality on the both the judging platform and official website.
- Built a backend system that connects with Google Forms to automatically extract participant emails addresses and send confirmation and ticket emails.

Extracurriculars

- Co-founder and Tech Lead for Machine Learning Student Network. — **2024 – 2025**
- Technical Director at Google Developer Student Club — **2024 – 2025**